

CampusEats — Assignment 2

1. Capabilities

- 1. Manage Accounts
- 2. Browse Restaurants
- 3. Browse Menus
- 4. Manage Cart
- 5. Place Orders
- 6. Process Payments

2. Service Design

Service	Responsible	Owns
Accounts Service	Harsh	users, addresses
Catalogue Service	Sushant	restaurants, menu_items
Orders Service	Harsh	carts, orders, order_items
Payments Service	Sushant	payments

3. Service Contracts

Accounts Service

Operation	Input	Output	Errors
createAccount	name, email, password	userId, status	email already exists
getAccount	userId	user details	user not found
updateAccount	userId, profile details	updated account	user not found

Catalogue Service

Operation	Input	Output	Errors
listRestaurants	area / filter	restaurant list	no restaurants found
getMenu	restaurantId	menu items and prices	restaurant not found
getMenuItem	itemId	item details	item not found

Orders Service

Operation	Input	Output	Errors
addToCart	userId, itemId, quantity	updated cart	item unavailable
getCart	userId	cart details	cart not found
placeOrder	userId, items, deliveryAddressId, paymentMethod	orderId, status, total	empty cart, item unavailable, payment failed, invalid address

Payments Service

Operation	Input	Output	Errors
makePayment	orderId, amount, paymentMethodId	paymentId, status	payment failed
getPayment	paymentId	payment details	payment not found

4. placeOrder Specification

Inputs

- studentId
- items: [{itemId, quantity}]
- deliveryAddressId
- paymentMethodId

Outputs

- orderId
- status: placed
- total
- estimatedMinutes

Error Cases

- item unavailable
- payment declined
- invalid address
- empty cart

Hidden from Callers

- Internal orders data and database structure
- How IDs are generated
- Which payment gateway is used
- Internal payment processing details
- Internal order processing logic

5. Service Validation

Service	Reachable	Self-contained	Has Contract	Independent	Loosely Coupled
Accounts Service	Yes	Yes	Yes	Yes	Yes
Catalogue Service	Yes	Yes	Yes	Yes	Yes
Orders Service	Yes	Yes	Yes	Yes	Yes
Payments Service	Yes	Yes	Yes	Yes	Yes

- Accounts Service: Owns user and address data and exposes account operations through its contract.
- Catalogue Service: Owns restaurant and menu data and provides restaurant and menu operations through its contract.
- Orders Service: Owns carts and orders and communicates with other services through published operations.
- Payments Service: Owns payment data and handles payment operations independently.
- No service directly accesses another service's tables.

- Each service has its own data ownership and published contract.